

Solution Requirements - OptiCrop

Project: OptiCrop – Smart Agricultural Production Optimization Engine

Functional Requirements

S.No	Category	Description	Priority
1	Authentication	Farmer/Admin login using secure credentials.	High
2	Authorization	Farmer views recommendations; Admin manages datasets/models.	High
3	External Interfaces	Weather API, Soil dataset, ML model.	High
4	Transactions	Accept soil inputs and generate crop recommendations.	High
5	Reporting	Recommendation report and prediction history.	Medium
6	Business Rules	Recommend crops based on soil nutrients and weather.	High
7	Compliance	Protect user data and follow privacy standards.	Medium
8	Other	Fertilizer suggestions and future crop support.	Medium

Non-Functional Requirements

S.No	Category	Description	Acceptance
1	Performance	Prediction generated quickly.	<2 sec
2	Scalability	Handles many users.	1000+ users
3	Security	Encrypted user data.	HTTPS & hashed passwords
4	Reliability	High availability.	99% uptime
5	Usability	Simple farmer-friendly UI.	Easy navigation
6	Other	Maintainable and portable.	Runs on Windows/Linux